

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Electrical Engineering

2. Subject Name: VLSI Circuits and Systems Laboratory L-T-P: 0-0-3 Credits:2

3. Pre-requisites:

New: Analog Electronics Laboratory

New: Digital Electronics Laboratory

4. Syllabus and reference books:

Syllabus:

As given in the lecture wise breakup.

Reference Books:

- 1) CMOS VLSI Design: A Circuits and Systems Perspective - by Weste & Harris (4th edition)
- 2) Digital Integrated Circuits by Rabaey, Chandrakasan, Nikolic (2nd edition)
- 3) Analog Integrated Circuit Design, (Second Edition) by Tony Chan Carusone, David Johns, Kenneth Martin, Wiley
- 4) CMOS Analog Circuit Design by Allen and Holberg, Oxford Indian Edition
- 5) Design of Analog CMOS Integrated Circuits by Behzad Razavi, Tata McGraw-Hill
- 6) Power Management Integrated Circuits Architecture, Design and Implementation, Amit Patra, Shailendra Baranwal, Ashis Maity, Samiran Dam, Syed Asif Eqbal Taylor & Francis Ltd, 2024

5. Lecture-wise break-up:

Sl. No.	Topics	No. of lectures
1	• Use of EDA tools and Analog VLSI Design Flow(DC, AC, Transient, Operating Point, Noise Analysis) • Use of EDA tools and digital VLSI design flow.	6
2	Digital Design: • Using HDL to design, simulate and synthesize combinational circuits. • Design of full adder. • Design, simulation and synthesis of sequential circuits.	9

	Design of up/down counters.	
3	• Design of different configurations of single-stage amplifiers.	6
4	• Design of current mirror topologies (accuracy, headroom, output impedance)	3
5	• Design of single-stage differential amplifier • Design of a folded cascode amplifier	6
6	• Design of a two-stage operational amplifier	6
7	• Design of PVT independent voltage reference	3
8	• Design of ring oscillator (Variation of frequency with PVT)	3
	Total Hours	42